Program 1

In [6]:

```python
import pandas as pd
import numpy as np
data = pd.read_csv('lab1.csv')
data
features = np.array(data)[:,:-1]
features
target = np.array(data)[:,-1]
target

for i ,val in enumerate(target):
    if val == 'yes':
        specific_h = features[i].copy()
        break

print(specific_h)

for i,val in enumerate(features):
    if target[i] == 'yes':
        for x in range(len(specific_h)):
            if val[x] != specific_h[x]:
                specific_h[x] = '?'
print(specific_h)
```

```
['sunny' 'warm' 'normal' 'strong' 'warm' 'same']
['sunny' 'warm' '?' 'strong' '?' '?']
```

Program 2

In [2]:

```
import numpy as np
import pandas as pd
data = pd.read_csv('lab1.csv')
data
features=np.array(data)[:,:-1]
features
target=np.array(data)[:,-1]
target
specific_h=features[0].copy()
print("initialization of specific_h an general_h")
print(specific_h)
general_h=[["?" for i in range(len(specific_h))]for i in range(len(specifi
print(general_h)
for i,h in enumerate(features):
    if target[i] =="yes":
        for x in range(len(specific_h)):

            if h[x]!=specific_h[x]:
                specific_h[x]='?'
                general_h[x][x]='?'
    if target[i]=="no":
        for x in range(len(specific_h)):
            if h[x]!=specific_h[x]:
                general_h[x][x]=specific_h[x]
            else:
                general_h[x][x]='?'
print(specific_h,"\n")
print(general_h,"\n")
indices=[i for i,val in enumerate(general_h) if val==['?','?','?','?','?',
for i in indices:
    general_h.remove(['?','?','?','?','?','?'])
print("\nfinal specific_h:",specific_h,sep="\n")
print("final general_h:",general_h,sep="\n")
```

```
initialization of specific_h an general_h
['sunny' 'warm' 'normal' 'strong' 'warm' 'same']
[['?', '?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?', '?',
'?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?', '?', '?', '?',
'?', '?'], ['?', '?', '?', '?', '?', '?']]
['sunny' 'warm' '?' 'strong' '?' '?']

[['sunny', '?', '?', '?', '?', '?'], ['?', 'warm', '?', '?', '?', '?'], ['?',
'?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?', '?', '?',
'?', '?', '?'], ['?', '?', '?', '?', '?', '?']]


final specific_h:
['sunny' 'warm' '?' 'strong' '?' '?']
final general_h:
[['sunny', '?', '?', '?', '?', '?'], ['?', 'warm', '?', '?', '?', '?']]
```

Program 3

In [ ]:

```
import pandas as pd
from collections import Counter
import math

tennis= pd.read_csv('tennis.csv')
print("\n given play tennis dataset:\n \n",tennis)

def entropy(alist):
    c = Counter(x for x in alist)
    instances = len(alist)
    prob = [x / instances for x in c.values()]
    return sum( [-p*math.log(p,2) for p in prob] )

def information_gain(d,split,target):
    splitting=d.groupby(split)
    n=len(d.index)
    agent = splitting.agg({target : [entropy,lambda x: len(x)/n ] })[targe
    agent.columns = ['entropy','observations']
    newentropy = sum(agent['entropy']*agent['observations'])
    oldentropy=entropy(d[target])
    return oldentropy - newentropy

def id3(sub,target,a):
    count = Counter(x for x in sub[target])#class of YES / NO
    if len(count)==1:
        return next(iter(count))#next i/p dataset
    else:
        gain=[information_gain(sub,attr,target) for attr in a]
        print("\n gain=",gain)
        maximum = gain.index(max(gain))
        best = a[maximum]
        print("\nbest attribute=",best)
        tree={best:{}}
        remaining=[i for i in a if i != best]

        for val,subset in sub.groupby(best):
            subtree = id3(subset,target,remaining)
            tree[best][val] = subtree
        return tree

names = list(tennis.columns)
print("\nlist of attributes:",names)
names.remove("PlayTennis")
print("\npredicting attributes:",names)
tree=id3(tennis,'PlayTennis',names)
print("\n\n the resultant decision tree is:\n")
print(tree)
```

Program 4

In [4]:

```python
import numpy as np
x=np.array(([2,9],[1,5],[3,6]),dtype=float)
y=np.array(([92],[86],[89]),dtype=float)
x=x/np.amax(x,axis=0)
y=y/100

def sigmoid(x):
    return 1/(1+np.exp(-x))
def derivatives_sigmoid(x):
    return x*(1-x)
epoch=7000
lr=0.1
inputlayer_neurons=2
hiddenlayer_neurons=3
output_neurons=1

wh=np.random.uniform(size=(inputlayer_neurons,hiddenlayer_neurons))
bh=np.random.uniform(size=(1,hiddenlayer_neurons))
wout=np.random.uniform(size=(hiddenlayer_neurons,output_neurons))
bout=np.random.uniform(size=(1,output_neurons))

for i in range(epoch):
    hinp1=np.dot(x,wh)
    hinp=hinp1+bh
    hlayer_act=sigmoid(hinp)
    outinp1=np.dot(hlayer_act,wout)
    outinp=outinp1+bout
    output=sigmoid(outinp)

    EO=y-output
    outgrad=derivatives_sigmoid(output)
    d_output=EO*outgrad
    EH=d_output.dot(wout.T)
    hiddengrad=derivatives_sigmoid(hlayer_act)
    d_hiddenlayer=EH*hiddengrad
    wout+=hlayer_act.T.dot(d_output)*lr

print("Input:\n"+str(x))
print("Actual output: \n"+str(y))
print("Predicted output:\n",output)
```

```
Input:
[[0.66666667 1.        ]
 [0.33333333 0.55555556]
 [1.         0.66666667]]
Actual output:
[[0.92]
 [0.86]
 [0.89]]
Predicted output:
 [[0.89652276]
 [0.87954289]
 [0.8931647 ]]
```

Program 5

In [28]:

```
import pandas as pd
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import MultinomialNB
from sklearn.metrics import accuracy_score
from sklearn import metrics

data=pd.read_csv('textdata12.csv',names=['message','label'])
print('the dataset is',data)
print('the dimensions of the dataset',data.shape)
data['labelnum']=data.label.map({'pos':1,'neg':0})
x=data.message
y=data.labelnum
print(x,y)

vectorizer=TfidfVectorizer()
data=vectorizer.fit_transform(x)
print("\n the feature of dataset:\n")
df=pd.DataFrame(data.toarray(),columns=vectorizer.get_feature_names_out())
df.head()
print("\n train test split")
xtrain,xtest,ytrain,ytest=train_test_split(data,y,test_size=0.3,random_sta
print("\n the total number of training data:",ytrain.shape)
print("\n the total number of test data:",ytest.shape)
clf=MultinomialNB().fit(xtrain,ytrain)
predict=clf.predict(xtest)
predicted=clf.predict(xtest)
print("\n accuracy of the classifier is",metrics.accuracy_score(ytest,pred
print("\n confusion matrix is\n",metrics.confusion_matrix(ytest,predicted)
print("\n classification report is\n",metrics.classification_report(ytest,
print("\n the value of precision is\n",metrics.precision_score(ytest,predi
print("\n the value of recall  is\n",metrics.recall_score(ytest,predicted)
```

```
the dataset is                                     message label
0                        i love sandwitch   pos
1                this is an amazing place   pos
2      i feel very good about these beers   pos
3                    this is my best work   pos
4                    what an awesome view   pos
5            i do not like this resturant   neg
6                i am tired of this stuff   neg
7                   i cant deal with this   neg
8                    he is my sworn enemy   neg
9                     my boss is horrible   neg
10               this is an awesome place   pos
11  i do not like the taste of this juice   neg
12                        i love to dance   pos
13      i am sick and tired of this place   neg
14                   what a great holiday   pos
15           that is bad locality to stay   neg
16         we will have good fun tomorrow   pos
17       i went to my enemy's house today   neg
the dimensions of the dataset (18, 2)
0                           i love sandwitch
1                  this is an amazing place
2        i feel very good about these beers
3                      this is my best work
4                      what an awesome view
5              i do not like this resturant
6                  i am tired of this stuff
7                     i cant deal with this
8                      he is my sworn enemy
9                       my boss is horrible
10                 this is an awesome place
11    i do not like the taste of this juice
12                          i love to dance
13        i am sick and tired of this place
14                     what a great holiday
15             that is bad locality to stay
16           we will have good fun tomorrow
17         i went to my enemy's house today
Name: message, dtype: object 0     1
1     1
2     1
3     1
4     1
5     0
6     0
7     0
8     0
9     0
10    1
11    0
12    1
13    0
14    1
15    0
16    1
17    0
Name: labelnum, dtype: int64
```

```
the feature of dataset:


train test split

the total number of training data: (12,)

the total number of test data: (6,)

accuracy of the classifier is 0.8333333333333334

confusion matrix is
[[3 0]
[1 2]]

classification report is
              precision    recall  f1-score   support

           0       0.75      1.00      0.86         3
           1       1.00      0.67      0.80         3

    accuracy                           0.83         6
   macro avg       0.88      0.83      0.83         6
weighted avg       0.88      0.83      0.83         6


the value of precision is
1.0

the value of recall  is
0.6666666666666666
```

Program 6

In [30]:

```
import warnings
import pandas as pd
col=['Age','Gender','Familylist','Diet','LifeStyle','Cholesterol','heartDi
data=pd.read_csv('lab6.csv',names=col)
print(data)
from sklearn.preprocessing import LabelEncoder
encoder=LabelEncoder()
for i in range(len(col)):
    data.iloc[:,i]=encoder.fit_transform(data.iloc[:,i])
X=data.iloc[:,0:6]
y=data.iloc[:,-1]
from sklearn.model_selection import train_test_split
X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.2)
from sklearn.naive_bayes import GaussianNB
clf=GaussianNB()
clf.fit(X_train,y_train)
y_pred=clf.predict(X_test)
from sklearn.metrics import confusion_matrix
print('confusion matrix',confusion_matrix(y_test,y_pred))
```

```
                   Age  Gender Familylist    Diet LifeStyle Cholesterol  \
0   SuperSeniorCitizen    Male        Yes  Medium  Sedetary        High
1   SuperSeniorCitizen  Female        Yes  Medium  Sedetary        High
2        SeniorCitizen    Male         No    High  Moderate  Borderline
3                 Teen    Male        Yes  Medium  Sedetary      Normal
4                Youth  Female        Yes    High   Athlete      Normal
5           MiddleAged    Male        Yes  Medium    Active        High
6                 Teen    Male        Yes    High  Moderate        High
7   SuperSeniorCitizen    Male        Yes  Medium  Sedetary        High
8                Youth  Female        Yes    High   Athlete      Normal
9        SeniorCitizen  Female         No    High   Athlete      Normal
10                Teen  Female         No  Medium  Moderate        High
11                Teen    Male        Yes  Medium  Sedetary      Normal
12          MiddleAged  Female         No    High   Athlete        High
13          MiddleAged    Male        Yes  Medium    Active        High
14               Youth  Female        Yes    High   Athlete  Borderline
15  SuperSeniorCitizen    Male        Yes    High   Athlete      Normal
16       SeniorCitizen  Female         No  Medium  Moderate  Borderline
17                Teen    Male        Yes  Medium  Sedetary      Normal

   heartDisease
0           Yes
1           Yes
2           Yes
3            No
4            no
5           Yes
6           Yes
7           Yes
8            No
9           Yes
10          Yes
11           No
12           No
13          Yes
14           No
15          Yes
16          Yes
17           No
confusion matrix [[1 0 0]
 [1 1 0]
 [1 0 0]]
```

```
C:\Users\BITM\AppData\Local\Temp\ipykernel_8384\3744051054.py:9: DeprecationW
arning: In a future version, `df.iloc[:, i] = newvals` will attempt to set th
e values inplace instead of always setting a new array. To retain the old beh
avior, use either `df[df.columns[i]] = newvals` or, if columns are non-uniqu
e, `df.isetitem(i, newvals)`
  data.iloc[:,i]=encoder.fit_transform(data.iloc[:,i])
C:\Users\BITM\AppData\Local\Temp\ipykernel_8384\3744051054.py:9: DeprecationW
arning: In a future version, `df.iloc[:, i] = newvals` will attempt to set th
e values inplace instead of always setting a new array. To retain the old beh
avior, use either `df[df.columns[i]] = newvals` or, if columns are non-uniqu
e, `df.isetitem(i, newvals)`
  data.iloc[:,i]=encoder.fit_transform(data.iloc[:,i])
C:\Users\BITM\AppData\Local\Temp\ipykernel_8384\3744051054.py:9: DeprecationW
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e values inplace instead of always setting a new array. To retain the old beh
avior, use either `df[df.columns[i]] = newvals` or, if columns are non-uniqu
e, `df.isetitem(i, newvals)`
  data.iloc[:,i]=encoder.fit_transform(data.iloc[:,i])
C:\Users\BITM\AppData\Local\Temp\ipykernel_8384\3744051054.py:9: DeprecationW
arning: In a future version, `df.iloc[:, i] = newvals` will attempt to set th
e values inplace instead of always setting a new array. To retain the old beh
avior, use either `df[df.columns[i]] = newvals` or, if columns are non-uniqu
e, `df.isetitem(i, newvals)`
  data.iloc[:,i]=encoder.fit_transform(data.iloc[:,i])
C:\Users\BITM\AppData\Local\Temp\ipykernel_8384\3744051054.py:9: DeprecationW
arning: In a future version, `df.iloc[:, i] = newvals` will attempt to set th
e values inplace instead of always setting a new array. To retain the old beh
avior, use either `df[df.columns[i]] = newvals` or, if columns are non-uniqu
e, `df.isetitem(i, newvals)`
  data.iloc[:,i]=encoder.fit_transform(data.iloc[:,i])
C:\Users\BITM\AppData\Local\Temp\ipykernel_8384\3744051054.py:9: DeprecationW
arning: In a future version, `df.iloc[:, i] = newvals` will attempt to set th
e values inplace instead of always setting a new array. To retain the old beh
avior, use either `df[df.columns[i]] = newvals` or, if columns are non-uniqu
e, `df.isetitem(i, newvals)`
  data.iloc[:,i]=encoder.fit_transform(data.iloc[:,i])
```
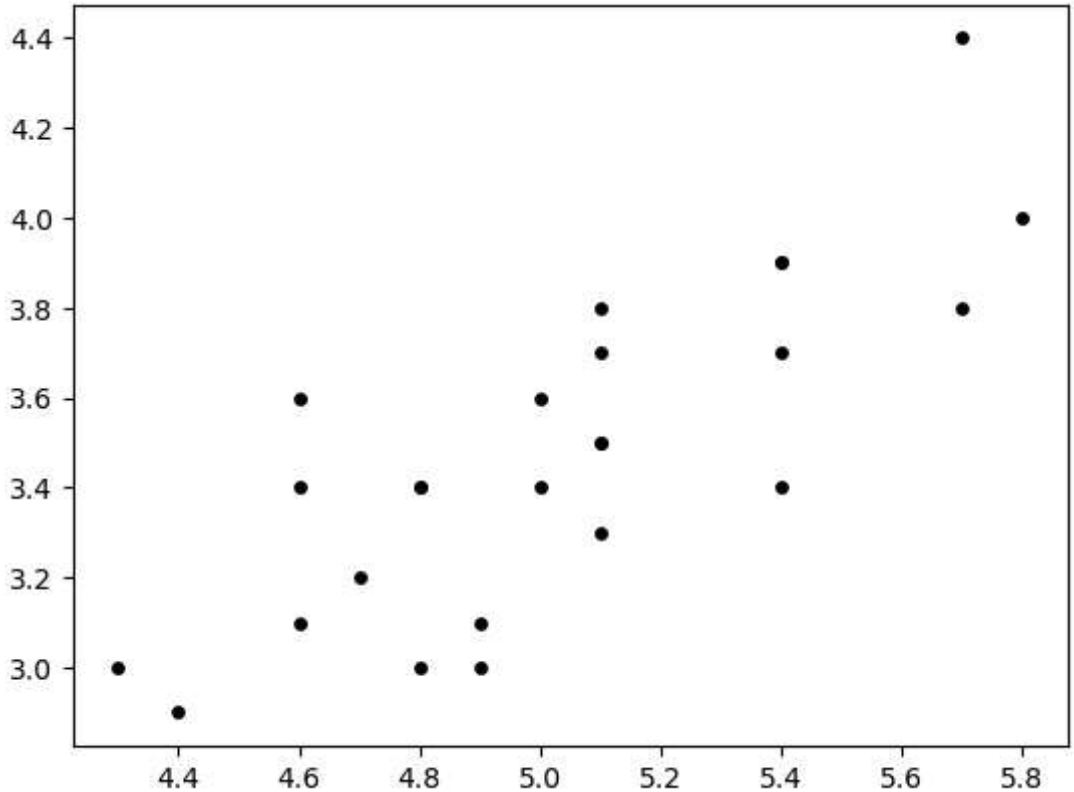

Program 7

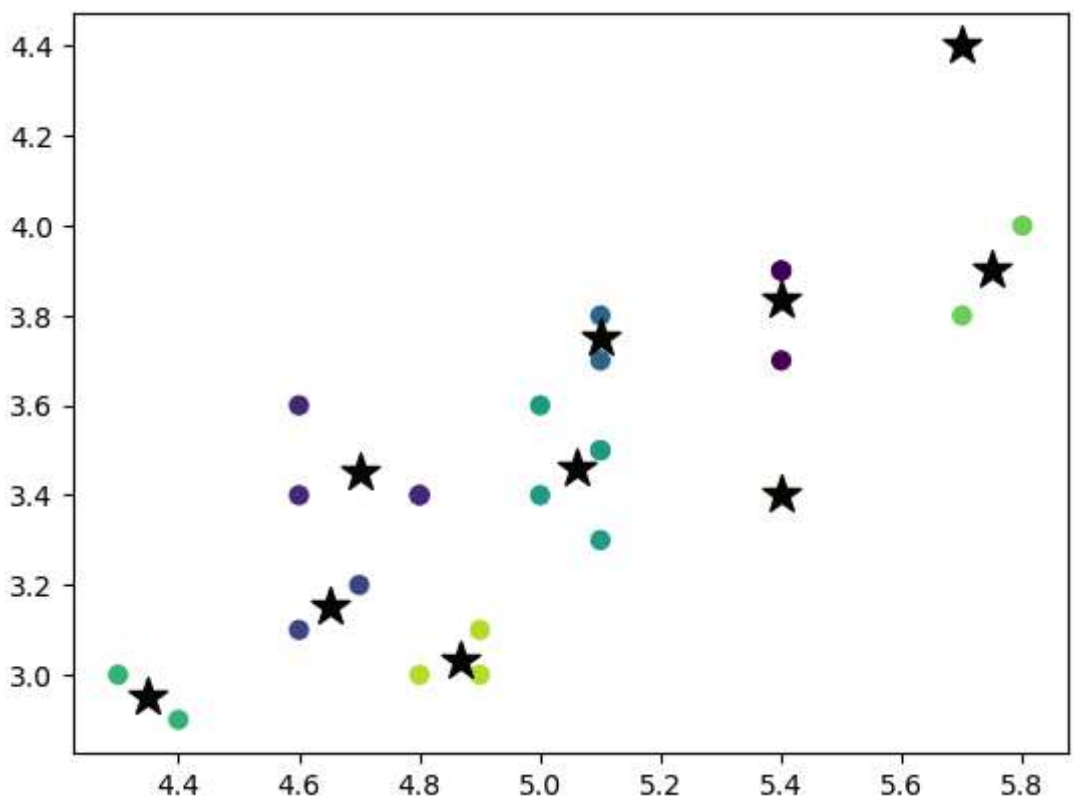
In [15]:

```
import numpy as np
import pandas as pd
from matplotlib import pyplot as plt
from sklearn.mixture import GaussianMixture
from sklearn.cluster import KMeans
data=pd.read_csv('lab7.csv')
print('Input Data and Shape')
print(data.shape)
data.head()
f1=data['V1'].values
f2=data['V2'].values
X=np.array(list(zip(f1,f2)))
print("X",X)
print("Graph for whole dataset")
plt.scatter(f1,f2,c='black',s=15)
plt.show()
kmeans=KMeans(10,random_state=42)
labels=kmeans.fit(X).predict(X)
print("labels ",labels)
centroids=kmeans.cluster_centers_
print("Centriods   ",centroids)
plt.scatter(X[:,0],X[:,1],c=labels,s=40,cmap='viridis')
print("Graph using Kmeans Algorithm ")
plt.scatter(centroids[:,0],centroids[:,1],marker='*',s=200,c='#050505')
plt.show()
gmm=GaussianMixture(n_components=3).fit(X)
labels=gmm.predict(X)
probs=gmm.predict_proba(X)
size=10*probs.max(1)**3
print("Graph using Em algorithm")
plt.scatter(X[:,0],X[:,1],c=labels,s=size,cmap='viridis')
plt.show()
```

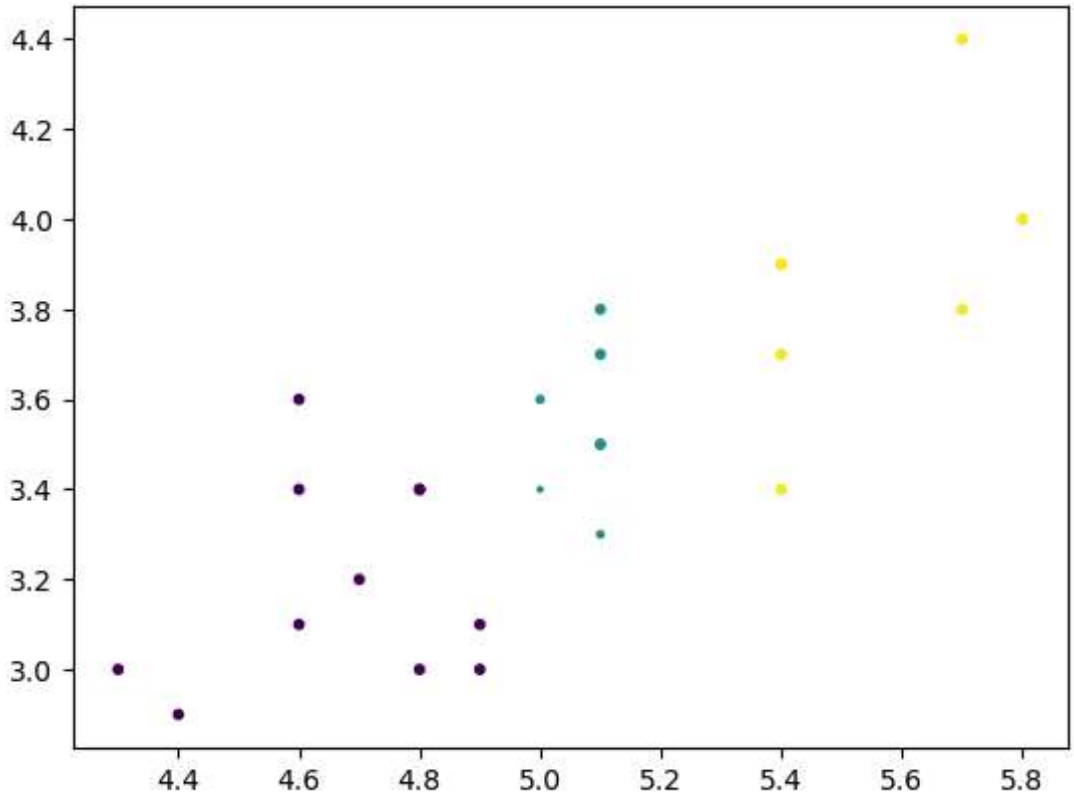
```
Input Data and Shape
(25, 2)
X [[5.1 3.5]
 [4.9 3. ]
 [4.7 3.2]
 [4.6 3.1]
 [5.  3.6]
 [5.4 3.9]
 [4.6 3.4]
 [5.  3.4]
 [4.4 2.9]
 [4.9 3.1]
 [5.4 3.7]
 [4.8 3.4]
 [4.8 3. ]
 [4.3 3. ]
 [5.8 4. ]
 [5.7 4.4]
 [5.4 3.9]
 [5.1 3.5]
 [5.7 3.8]
 [5.1 3.8]
 [5.4 3.4]
 [5.1 3.7]
 [4.6 3.6]
 [5.1 3.3]
 [4.8 3.4]]
Graph for whole dataset
```



```
C:\Users\BITM\Documents\anaconda3\Lib\site-packages\sklearn\cluster\_kmeans.p
y:1412: FutureWarning: The default value of `n_init` will change from 10 to
'auto' in 1.4. Set the value of `n_init` explicitly to suppress the warning
  super()._check_params_vs_input(X, default_n_init=10)
C:\Users\BITM\Documents\anaconda3\Lib\site-packages\sklearn\cluster\_kmeans.p
y:1436: UserWarning: KMeans is known to have a memory leak on Windows with MK
L, when there are less chunks than available threads. You can avoid it by set
ting the environment variable OMP_NUM_THREADS=1.
  warnings.warn(

labels  [5 8 2 2 5 0 1 5 6 8 0 1 8 6 7 4 0 5 7 3 9 3 1 5 1]
Centriods    [[5.4        3.83333333]
 [4.7        3.45       ]
 [4.65       3.15       ]
 [5.1        3.75       ]
 [5.7        4.4        ]
 [5.06       3.46       ]
 [4.35       2.95       ]
 [5.75       3.9        ]
 [4.86666667 3.03333333]
 [5.4        3.4        ]]
Graph using Kmeans Algorithm
```




```
Graph using Em algorithm

C:\Users\BITM\Documents\anaconda3\Lib\site-packages\sklearn\cluster\_kmeans.p
y:1436: UserWarning: KMeans is known to have a memory leak on Windows with MK
L, when there are less chunks than available threads. You can avoid it by set
ting the environment variable OMP_NUM_THREADS=1.
  warnings.warn(
```


4.4
4.2
4.0
3.8
3.6
3.4
3.2
3.0
4.4
4.6
4.8
5.0
5.2
5.4
5.6
5.8

Program 8

In [16]:

```python
import csv
import random
import math
import operator
def loadDataset(filename,split,trainingSet=[],testSet=[]):
    with open(filename) as csvfile:
        lines=csv.reader(csvfile)
        dataset=list(lines)
        for x in range(len(dataset)-1):
            for y in range(4):
                dataset[x][y]=float(dataset[x][y])
            if random.random()<split:
                trainingSet.append(dataset[x])
            else:
                testSet.append(dataset[x])
def euclideanDistance(instance1,instance2,length):
    distance=0
    for x in range(length):
        distance+=pow((instance1[x]-instance2[x]),2)
    return math.sqrt(distance)
def getNeighbors(trainingSet,testInstance,k):
    distances=[]
    length=len(testInstance)-1
    for x in range(len(trainingSet)):
        dist=euclideanDistance(testInstance,trainingSet[x],length)
        distances.append((trainingSet[x],dist))
    distances.sort(key=operator.itemgetter(1))
    neighbors=[]
    for x in range(k):
        neighbors.append(distances[x][0])
    return neighbors
def getResponse(neighbors):
    classVotes={}
    for x in range(len(neighbors)):
        response=neighbors[x][-1]
        if response in classVotes:
            classVotes[response]+=1
        else:
            classVotes[response]=1
    sortedVotes=sorted(classVotes.items(),key=operator.itemgetter(1),rever
    return sortedVotes[0][0]
def getAccuracy(testSet,predictions):
    correct=0
    for x in range(len(testSet)):
        if testSet[x][-1]==predictions[x]:
            correct+=1
    return (correct/float(len(testSet)))*100.0
def main():
    trainingSet=[]
    testSet=[]
    split=0.67
    loadDataset('iris.csv',split,trainingSet,testSet)
    print('\n Number of Training data:'+(repr(len(trainingSet))))
    print("Number of test data :"+repr(len(testSet)))
    predictions=[]
    k=3
    print("\n the predictions are: ")
```

```
    for x in range(len(testSet)):
        neighbors=getNeighbors(trainingSet,testSet[x],k)
        result=getResponse(neighbors)
        predictions.append(result)
        print('predicted= '+repr(result)+',actual='+repr(testSet[x][-1]))
    accuracy=getAccuracy(testSet,predictions)
    print("\n The accuracy is "+repr(accuracy)+'%')
main()
```

```
 Number of Training data:91
Number of test data :58

 the predictions are:
predicted= 'Iris-setosa',actual='Iris-setosa'
predicted= 'Iris-setosa',actual='Iris-setosa'
predicted= 'Iris-setosa',actual='Iris-setosa'
predicted= 'Iris-setosa',actual='Iris-setosa'
predicted= 'Iris-setosa',actual='Iris-setosa'
predicted= 'Iris-setosa',actual='Iris-setosa'
predicted= 'Iris-setosa',actual='Iris-setosa'
predicted= 'Iris-setosa',actual='Iris-setosa'
predicted= 'Iris-setosa',actual='Iris-setosa'
predicted= 'Iris-setosa',actual='Iris-setosa'
predicted= 'Iris-setosa',actual='Iris-setosa'
predicted= 'Iris-setosa',actual='Iris-setosa'
predicted= 'Iris-setosa',actual='Iris-setosa'
predicted= 'Iris-setosa',actual='Iris-setosa'
predicted= 'Iris-setosa',actual='Iris-setosa'
predicted= 'Iris-setosa',actual='Iris-setosa'
predicted= 'Iris-setosa',actual='Iris-setosa'
predicted= 'Iris-setosa',actual='Iris-setosa'
predicted= 'Iris-setosa',actual='Iris-setosa'
predicted= 'Iris-setosa',actual='Iris-setosa'
predicted= 'Iris-versicolor',actual='Iris-versicolor'
predicted= 'Iris-versicolor',actual='Iris-versicolor'
predicted= 'Iris-versicolor',actual='Iris-versicolor'
predicted= 'Iris-versicolor',actual='Iris-versicolor'
predicted= 'Iris-versicolor',actual='Iris-versicolor'
predicted= 'Iris-versicolor',actual='Iris-versicolor'
predicted= 'Iris-versicolor',actual='Iris-versicolor'
predicted= 'Iris-virginica',actual='Iris-versicolor'
predicted= 'Iris-versicolor',actual='Iris-versicolor'
predicted= 'Iris-versicolor',actual='Iris-versicolor'
predicted= 'Iris-versicolor',actual='Iris-versicolor'
predicted= 'Iris-versicolor',actual='Iris-versicolor'
predicted= 'Iris-versicolor',actual='Iris-versicolor'
predicted= 'Iris-versicolor',actual='Iris-versicolor'
predicted= 'Iris-versicolor',actual='Iris-versicolor'
predicted= 'Iris-versicolor',actual='Iris-versicolor'
predicted= 'Iris-versicolor',actual='Iris-versicolor'
predicted= 'Iris-versicolor',actual='Iris-versicolor'
predicted= 'Iris-versicolor',actual='Iris-versicolor'
predicted= 'Iris-versicolor',actual='Iris-versicolor'
predicted= 'Iris-virginica',actual='Iris-virginica'
predicted= 'Iris-virginica',actual='Iris-virginica'
predicted= 'Iris-versicolor',actual='Iris-virginica'
predicted= 'Iris-virginica',actual='Iris-virginica'
predicted= 'Iris-virginica',actual='Iris-virginica'
predicted= 'Iris-virginica',actual='Iris-virginica'
predicted= 'Iris-virginica',actual='Iris-virginica'
predicted= 'Iris-virginica',actual='Iris-virginica'
predicted= 'Iris-virginica',actual='Iris-virginica'
predicted= 'Iris-virginica',actual='Iris-virginica'
predicted= 'Iris-virginica',actual='Iris-virginica'
```

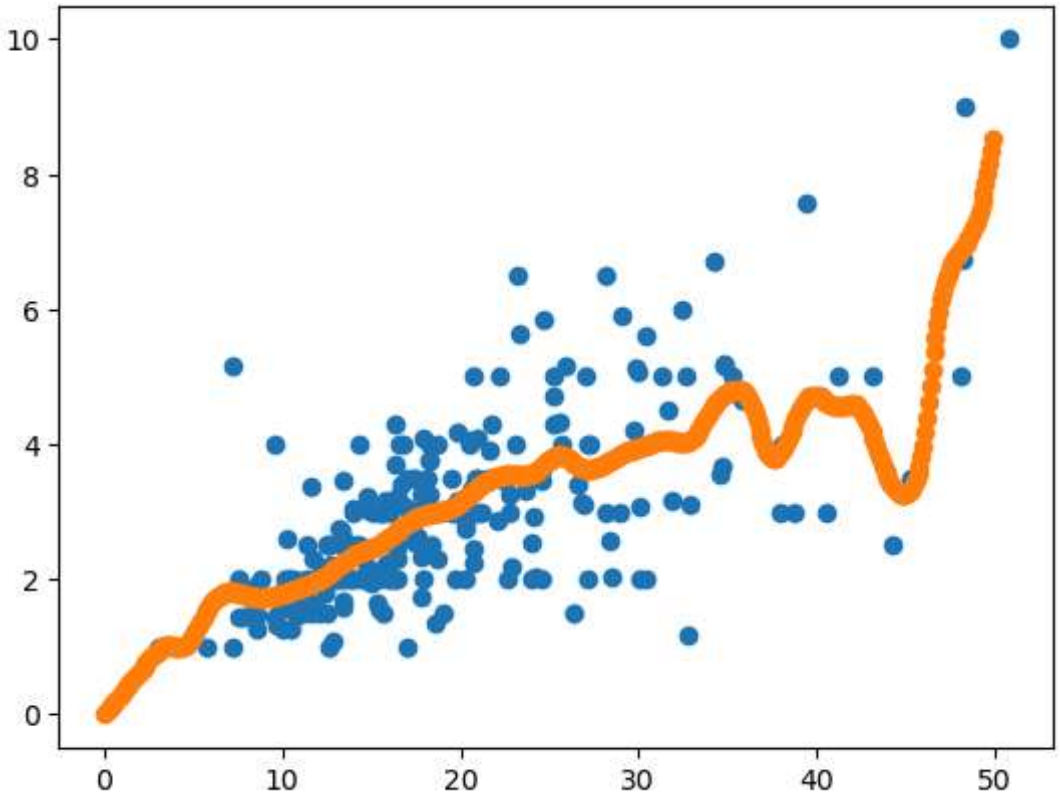
```
predicted= 'Iris-virginica',actual='Iris-virginica'
predicted= 'Iris-virginica',actual='Iris-virginica'
predicted= 'Iris-virginica',actual='Iris-virginica'
predicted= 'Iris-virginica',actual='Iris-virginica'
predicted= 'Iris-virginica',actual='Iris-virginica'
predicted= 'Iris-virginica',actual='Iris-virginica'

 The accuracy is 96.55172413793103%
```

Program 9

In [17]:

```python
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
tou=1
data=pd.read_csv("tips.csv")
X_train=np.array(data.total_bill)
print(X_train)
X_train=X_train[:,np.newaxis]
print(len(X_train))
y_train=np.array(data.tip)
X_test=np.array([i/10 for i in range(500)])
X_test=X_test[:,np.newaxis]
y_test=[]
count=0
for r in range(len(X_test)):
    wts=np.exp(-np.sum((X_train-X_test[r])**2,axis=1)/(2*tou**2))
    W=np.diag(wts)
    factor1=np.linalg.inv(X_train.T.dot(W).dot(X_train))#factor=XT.W.X
    parameters=factor1.dot(X_train.T).dot(W).dot(y_train)
    prediction=X_test[r].dot(parameters)
    y_test.append(prediction)
    count+=1
print(len(y_test))
y_test=np.array(y_test)
plt.plot(X_train.squeeze(),y_train,'o')
plt.plot(X_test.squeeze(),y_test,'o')
plt.show()
```

```
[16.99 10.34 21.01 23.68 24.59 25.29  8.77 26.88 15.04 14.78 10.27 35.26
 15.42 18.43 14.83 21.58 10.33 16.29 16.97 20.65 17.92 20.29 15.77 39.42
 19.82 17.81 13.37 12.69 21.7  19.65  9.55 18.35 15.06 20.69 17.78 24.06
 16.31 16.93 18.69 31.27 16.04 17.46 13.94  9.68 30.4  18.29 22.23 32.4
 28.55 18.04 12.54 10.29 34.81  9.94 25.56 19.49 38.01 26.41 11.24 48.27
 20.29 13.81 11.02 18.29 17.59 20.08 16.45  3.07 20.23 15.01 12.02 17.07
 26.86 25.28 14.73 10.51 17.92 27.2  22.76 17.29 19.44 16.66 10.07 32.68
 15.98 34.83 13.03 18.28 24.71 21.16 28.97 22.49  5.75 16.32 22.75 40.17
 27.28 12.03 21.01 12.46 11.35 15.38 44.3  22.42 20.92 15.36 20.49 25.21
 18.24 14.31 14.    7.25 38.07 23.95 25.71 17.31 29.93 10.65 12.43 24.08
 11.69 13.42 14.26 15.95 12.48 29.8   8.52 14.52 11.38 22.82 19.08 20.27
 11.17 12.26 18.26  8.51 10.33 14.15 16.   13.16 17.47 34.3  41.19 27.05
 16.43  8.35 18.64 11.87  9.78  7.51 14.07 13.13 17.26 24.55 19.77 29.85
 48.17 25.   13.39 16.49 21.5  12.66 16.21 13.81 17.51 24.52 20.76 31.71
 10.59 10.63 50.81 15.81  7.25 31.85 16.82 32.9  17.89 14.48  9.6  34.63
 34.65 23.33 45.35 23.17 40.55 20.69 20.9  30.46 18.15 23.1  15.69 19.81
 28.44 15.48 16.58  7.56 10.34 43.11 13.   13.51 18.71 12.74 13.   16.4
 20.53 16.47 26.59 38.73 24.27 12.76 30.06 25.89 48.33 13.27 28.17 12.9
 28.15 11.59  7.74 30.14 12.16 13.42  8.58 15.98 13.42 16.27 10.09 20.45
 13.28 22.12 24.01 15.69 11.61 10.77 15.53 10.07 12.6  32.83 35.83 29.03
 27.18 22.67 17.82 18.78]
244
500
```

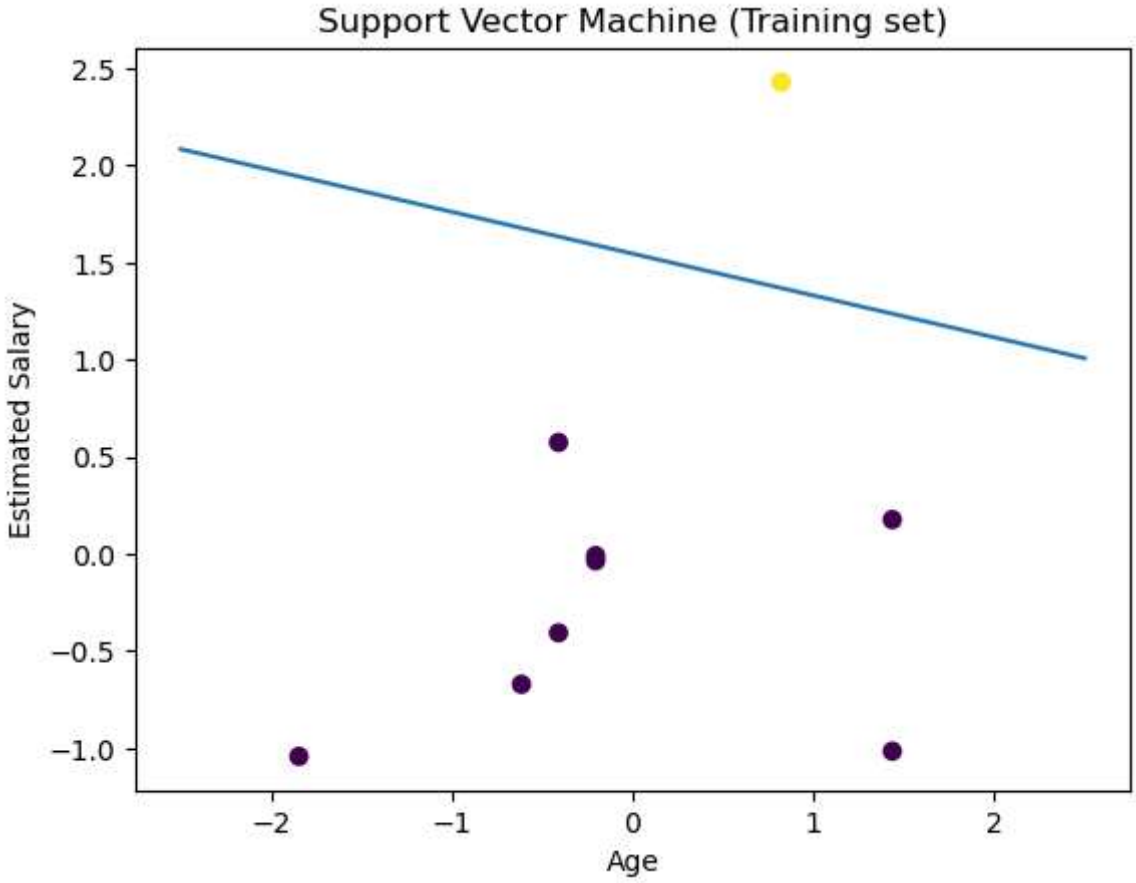

10
8
6
4
2
0
0
10
20
30
40
50

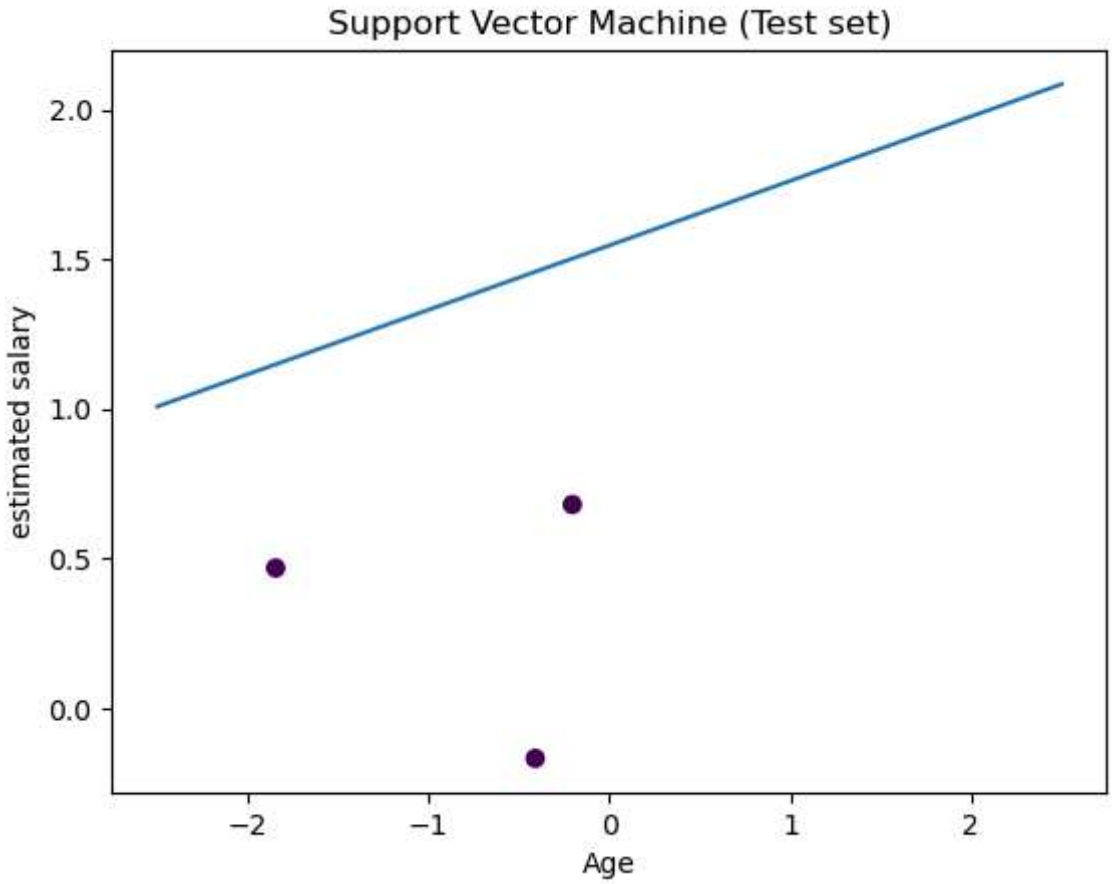
Program 10

In [3]:

```
#implement and demonstrate classification algorithm using support vector m
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
datasets = pd.read_csv('lab10.csv')
X = datasets.iloc[:, [2,3]].values
Y = datasets.iloc[:, 4].values
from sklearn.model_selection import train_test_split
X_Train, X_Test, Y_Train, Y_Test = train_test_split(X, Y, test_size = 0.25
random_state = 0)
from sklearn.preprocessing import StandardScaler
sc_X = StandardScaler()
X_Train = sc_X.fit_transform(X_Train)
X_Test = sc_X.transform(X_Test)
from sklearn.svm import SVC
classifier = SVC(kernel = 'linear', random_state = 0)
classifier.fit(X_Train, Y_Train)
Y_Pred = classifier.predict(X_Test)
from sklearn import metrics
print("Accuracy score ",metrics.accuracy_score(Y_Test, Y_Pred))
plt.scatter(X_Train[:,0], X_Train[:, 1],c=Y_Train)
plt.title('Support Vector Machine (Training set)')
plt.xlabel('Age')
plt.ylabel('Estimated Salary')
w=classifier.coef_[0]
a=-w[0]/w[1]
xx=np.linspace(-2.5,2.5)
yy=a*xx -(classifier.intercept_[0])/w[1]
plt.plot(xx,yy)
plt.show();
plt.scatter(X_Test[:,0], X_Test[:, 1],c=Y_Test)
plt.title('Support Vector Machine (Test set)')
plt.xlabel('Age')
plt.ylabel('estimated salary')
w= classifier.coef_[0]
a=w[0]/w[1]
xx=np.linspace(-2.5,2.5)
yy=a*xx-(classifier.intercept_[0])/w[1]
plt.plot(xx,yy)
plt.show()

```

```
Accuracy score  1.0
```


Support Vector Machine (Training set)
Estimated Salary
Age
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−2
−1
0
1
2


Support Vector Machine (Test set)
estimated salary
Age
2.0
1.5
1.0
0.5
0.0
−2
−1
0
1
2

In [19]:

```

```

In [ ]:

```

```